RPSMaster6000: an algorithmic bot that exploits human bias

1st Abu Safwan Md Farhan (22201340)
Computer Science and Engineering (SDS)
Brac University
Dhaka, Bangladesh
abusafwan.md.farhan@g.bracu.ac.bd

2nd Golam Mostofa Mahin (21301564)
Computer Science and Engineering (SDS)
Brac University
Dhaka, Bangladesh
golam.mostofa.mahin@g.bracu.ac.bd

3rd Shabiba Binta Shahadat (22101907)
Computer Science and Engineering (SDS)
Brac University
Dhaka, Bangladesh
shabiba.binta.shahadat@g.bracu.ac.bd

4th Mahibi Islam (2201828)
Computer Science and Engineering (SDS)
Brac University
Dhaka, Bangladesh
mahibi.islam@g.bracu.ac.bd

***Abstract*—In this paper, we introduce RPSMaster6000: an algorithmic bot that generates moves that exploit human biases. Humans do not play rock-paper-scissors completely randomly: they have certain biases. This bot tries to exploit those biases and win.**

***Index Terms*—RPSMaster6000, bias, rock-paper-scissors, digital-logic**

I. Implementation

A. Background

RPSMaster6000 is fundamentally based on [1]. It is useful only because of the biases that humans possess. If a human wins with a move in a specific round, then it is likely that he will play that winning move again in the next round. Moreover, if a human loses, s/he tends to switch to the move that would beat the move that s/he just lost to. Using these insights, RPSMaster6000 generates its moves.

1) RPSMaster6000 will always play paper for the first move.
2) If the human wins with rock, RPSMaster6000 will play paper in the next round.
3) If the human wins with paper, RPSMaster6000 will play with scissors in the next round.
4) If the human wins with scissors, RPSMaster6000 will play rock in the next round.
5) If the human loses with rock, RPSMaster6000 will play rock in the next round.
6) If the human loses with paper, RPSMaster6000 will play paper in the next round.
7) If the human loses with scissors, RPSMaster6000 will play scissors in the next round.
8) In the case of a tie, no strategy is clearly advantageous. So, in the event of a tie, we assumed that the human will repeat the same move in the next round as well, and made RPSMaster6000 generate moves that would combat that. For example, if both players play rock, RPSMaster6000 will play scissors next.

B. Working Principle

The 74147 encoder encodes the moves of the human into binary (10 for rock, 11 for paper, 01 for scissors).

The 74151 MUX outputs 1 when the human wins and 0 for ties and losses. In other words, HW is HIGH when human wins and LOW otherwise.

The part of the circuit on the bottom right (XNORs gate and AND gate) computes whether RPSMaster6000 won or lost. In other words, BW is HIGH when RPSMaster6000 wins and LOW otherwise. From HW and BW, we can detect ties.

The 74154 decoder and the 74147 encoder on the top left together calculate RPSMaster6000's next move. The calculated moves are latched by two D Flip-flops.

Finally, RPSMaster6000's move is shown via two white LEDs.

C. Challenges faced

1) This project was very time consuming. We had to risk losing our grades in other courses. This type of project is probably not recommendable to future CSE350 course-takers. A much simpler project would have probably been better.
2) A lot of wires were neeeded. At one point, we had to go to the market just to buy jumper wires and a breadboard.
3) Working on this project was very stressful and frustrating due to the complexity of the circuit.

II. Conclusion

In the end, we were able to successfully get RPSMaster6000 to generate the correct moves. A list of potential extensions to this project is provided herein:

1) Implement a score counter
2) Make RPSMaster6000 capable to generate moves randomly too.

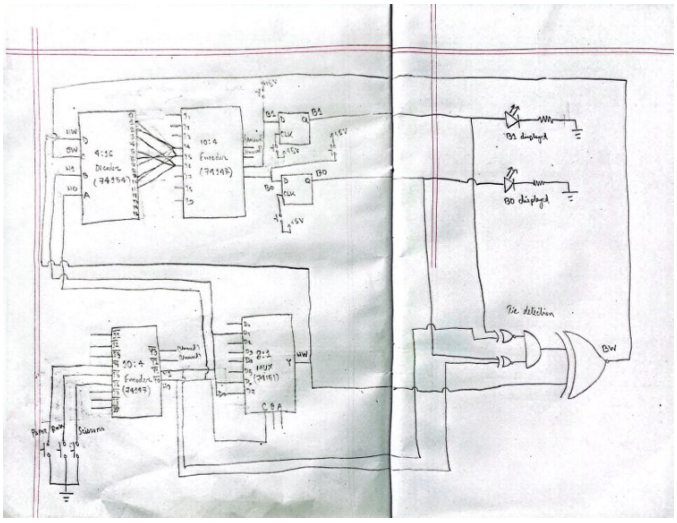


Fig. 1. RPSMaster6000 circuit diagram (simplified)

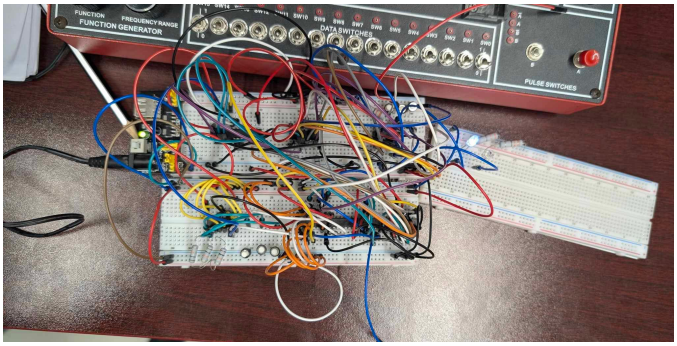


Fig. 2. RPSMaster6000 live demonstration

REFERENCES

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